

A virtual machine that you create using software like VirtualBox, VMWare Workstation / Player and Hyper-V have multiple components, some that are emulated, and some that are virtualised. For example the CD drive, the network card, and the floppy drive could be emulated, while the processor, sound card etc. are virtualised.

You might wonder though, why virtualise? Why don't we just find a way to run Linux on Windows without needing to create a virtual machine for it to run on? Here the answer is simple. Operating systems are not designed that way. Operating Systems are designed to be the thing that is in control of hardware. You could rewrite the operating system to cooperate and run within another operating system. Such a thing exists and is called paravirtualisation. However consider a closed source operating system like Windows. You cannot modify Windows to run cooperatively within Linux! Microsoft will have to add support for such a thing, and even then older versions of windows will not support it.

An advantage you might be able to see here with virtualisation is that it separates the software from the hardware even more. For instance, the OS running in a virtual machine need not care if the CD drive actually



Train simulator is a game that tries to capture the experience of driving a train through scenic routes.